
PROFESSOR

Julia Nolfo

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CLASS MEETINGS

Section 01

Tuesdays and Thursdays

4:35pm - 6:20pm

Harney 148

CLASS MEETINGS

Section 02

Tuesdays and Thursdays

6:30pm - 8:15pm

Harney 148

Introduction to Computer Science I

COMPUTER SCIENCE 110 | SPRING 2026

INTRODUCTION TO COMPUTER SCIENCE I is an introductory course in Computer Science at the University of San Francisco. Through the learning of the programming language **PYTHON**, students will use procedures, parameter passing, block structures, data types, arrays, abstract data structures, conditional control, iterative processes, and input/output in programming solutions to a variety of problems. Top-down and bottom-up design and functional decomposition to aid in the development of programs. This course is designed for students with very little or no programming experience. In this course, students will learn introductory-level programming concepts and techniques that will become a foundation for the future of their Computer Science education.

COURSE LEARNING OUTCOMES

- An introduction to the field of computer science.
- Learn the foundations of computer programming, computer algorithms, control structures, and data structures.
- Practice computer programming using the Python Programming Language.
- Learn how to perform numerical computations, process and transform text, and draw graphics and animations.

CORE B1 LEARNING OUTCOMES

In accordance with the Core B1 learning outcomes, through lecture, coding assignments, and quizzes CS 110 will teach you how to:

- **Design a mathematical (algorithmic) solution.**
You will learn how to design algorithmic solutions, using pseudo-code, for coding assignments.
- **Implement the design or identify and correct problems with the design.**
You will learn to implement (code) working programs for your designed solutions. You will also be given sample programs in lecture and will learn how to identify the pros and cons of each option.
- **Evaluate the validity of a solution and its relevance to the original problem using reasoned discourse as the norm for decision making.**
You will learn to explain and defend your reasoning for solving a problem in a particular way, gaining the ability to identify and fix bugs and refactor your code.

PROGRAM LEARNING OUTCOMES

- **Theory:** Explain and analyze standard computer science algorithms and describe and analyze theoretical aspects of various programming languages
- **Application:** Apply problem-solving skills to implement medium and large scale programs in a variety of programming languages
- **Systems:** Describe the interactions between low-level hardware, operating systems, and applications
- **Project:** Demonstrate effective communication and organization as part of a team of software developers or researchers collaborating on a large computer program

Required Materials

STARTING OUT WITH PYTHON

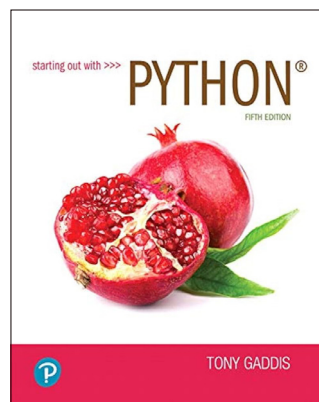
BY: TONY GADDIS

5th Edition

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Course Curriculum

This is a general outline for the semester. Please refer to Canvas for the most up-to-date lectures, assignments, and due dates.

UNIT 1: FOUNDATIONAL PROGRAMMING

- Week 1:** Introduction to the Course
- Week 2:** Input, Processing, and Output
- Week 3:** Decision Structures and Boolean Logic
- Week 4:** Repetition Structures
- Week 5:** Functions
- Week 6:** Turtle Graphics and the Python Math Module
- Week 7:** Pseudocode & How to Think Like a Computer Scientist
- Week 8:** Midterm Exam Review

MIDTERM EXAM

Week 9: Midterm Exam

UNIT 2: DATA STRUCTURES AND INTERMEDIATE PROGRAMMING

- Week 10:** Files & Exceptions
- Week 11:** Lists and Tuples
- Week 12:** More About Strings
- Week 13:** Dictionaries and Sets
- Week 14:** Dictionaries and Sets
- Week 15:** Final Exam Review

FINAL EXAM

Finals Week: Final Exam

Grading

GRADE BREAKDOWN

Setup	1%
Participation & Attendance	10%
Reading Assignments	0%
Homework Programs	12%
Individual Quizzes	25%
Team Quizzes	5%
Labs	21%
Midterm Exam	11%
Final Exam	15%

SETUP

At the start of the semester, students will complete a series of assignments to set up their environment for use throughout the course.

PARTICIPATION & ATTENDANCE

Attendance will be recorded at the start of each class period. Attendance reports are available to students on Canvas via Roll Call Attendance. Throughout the course, students will be required to participate in class discussions on Piazza and complete in-class programs during lecture.

READING ASSIGNMENTS

Each week, students will be assigned to read a chapter from *Starting Out with Python* by Tony Gaddis to supplement class lectures. However, students will not be graded on these assignments as the Weekly Quizzes will serve as a test of the student's comprehension of the reading as well as lectures.

HOMEWORK PROGRAMS

Each week, students will be assigned Homework Programs that focus on the concepts learned throughout the week in lecture, readings, and activities. Students will individually write a series of programs to demonstrate their knowledge of these concepts as well as any previously learned concepts. Homework Programs will be assigned at the end of each chapter.

INDIVIDUAL AND TEAM QUIZZES

Quizzes will take place at the beginning of each week during class. Quizzes will test the previous week's topics and consist of ten multiple-choice questions. All quizzes are "closed book" and will be taken on paper. Students will take the quiz individually, then immediately retake the same quiz with their assigned group.

LABS

Additional programming labs will be assigned throughout the course to give students more practice with the concepts learned in class. Labs will be more involved than Homework Programs, challenging students to write more complex programs using concepts learned in class. Labs will be completed in pairs during class.

MIDTERM EXAM

There will be a midterm exam about halfway through the semester, covering material from the first half of the semester. It will consist of two parts: multiple-choice and coding problems. The midterm will be taken individually during class.

FINAL EXAM

There will be a final exam at the end of the semester, during exam week. It will be similar to the midterm, with a multiple-choice part and a coding part, but will cover material from the whole course.

EXTRA CREDIT

Extra credit assignment will be available for optional completion throughout the course. See Course-Specific Policies for more information.

FINAL GRADES

Final grades for transcripts will come directly from a student's total grade on Canvas.

Final grades will be awarded as follows:

A+	96% and higher
A	≥ 93% and < 96%
A-	≥ 90% and < 93%
B+	≥ 87% and < 90%
B	≥ 83% and < 87%
B-	≥ 80% and < 83%
C+	≥ 77% and < 80%
C	≥ 73% and < 77%
C-	≥ 70% and < 73%
D+	≥ 67% and < 70%
D	≥ 63% and < 67%
D-	≥ 60% and < 63%
F	< 60%

Course-Specific Policies

ASSIGNMENT SUBMISSION DEADLINES

Unless otherwise specified, assignments are to be completed by students individually, and are due by 11:59pm on the due date. Late submissions will not be accepted unless arrangements have been made with the professor **prior** to the assignment's due date (See "Assignment Extensions" section below).

ASSIGNMENT EXTENSIONS

Each student is granted three (3) Assignment Extensions throughout the semester. Assignment extensions may be applied to Homework Programs. Assignment Extensions may **not** be applied to the Midterm Exam, Final Exam, Quizzes, Labs, or Extra Credit. If a student requests and is approved to utilize an assignment extension, they will be granted a 48 hour extension on the assignment. In order to use an Assignment Extension, the student must:

1. Email the Professor 24 hours before the assignment's deadline requesting an extension
2. The student must have used less than three (3) Assignment Extensions throughout the semester

Students are **not** to email Teacher's Assistants late assignments for grading. If a student does so, they will receive a zero on the assignment. All assignment extensions must be approved by the Professor via email. All assignment submissions must be made via Canvas.

ATTENDANCE

Attendance is mandatory. Attendance will be recorded every class meeting and is worth a small percentage of the student's final grade to encourage students to attend class for in-person instruction and assignments. If a student anticipates they will be absent from class for any reason, they should notify the instructor via email **prior to the start of class**. If this is done, the student will be excused from attendance that day (i.e. neither absent nor present) which does not affect their grade. Students will be marked absent if they do not notify the instructor prior to the start of class of their absence (i.e. "no call, no show"), affecting their grade.

MISSED QUIZZES AND IN-CLASS PROGRAMS

Makeups are not offered for quizzes or in-class programs. If a student misses a quiz or in-class program, they will either be excused from the quiz or in-class program if they are given an excused absence (see above) or receive a zero on the quiz or in-class program for an unexcused absence (see above) or tardiness to class. After 3 excused quizzes or in-class programs, the students will receive zeros on any further missed quizzes or in-class programs, regardless of whether their absence is excused or not.

MISSED LABS, MIDTERM EXAM, AND FINAL EXAM

Makeups are not offered for labs, the midterm exam, or the final exam. If a student is not present to complete a lab, the midterm exam, or the final exam, the student will receive a zero on the assignment, regardless of whether their absence is excused or not.

ACADEMIC HONESTY

All assignments are to be completed individually unless otherwise specified on the assignment. Students may help each other find and correct programming problems, but they may not share or copy each other's code. Academic dishonesty – e.g., copying another student's code – is **not** tolerated. Students are encouraged to post questions on Piazza or meet with a TA or the professor if they have questions regarding assignments or this policy.

Students may:

- Receive help from the professor or a TA
- Discuss with the professor, TA, or other students the requirements of assignments, bugs and general problems they are having making a program run properly, or high-level algorithms.

Students may NOT:

- Copy another student's code or solution to Homework Programs, Labs, Projects, Extra Credit, or Exams.
- Copy code from sources outside the course resources.
- Receive unapproved help from an outside source, including a tutor, a family member, or a friend.
- Submit code or program solutions which have, *in whole or in part*, been copied from any other source (including another student, online resources such as StackOverflow, Artificial Intelligence tools such as ChatGPT, or any another text) or written by anyone other than the student.

ChatGPT and AI Tools

Students may not utilize ChatGPT or any other AI tools to complete, in whole or in part, any assignments in this course. All concepts and materials needed to complete course assignments can be found in lecture slides and the textbook. If a student needs clarification on a topic or assistance with an assignment, they should attend the Professor's or Teacher's Assistant's office hours, post on Piazza, or visit the CS Tutoring Center.

Submitted code will be checked for evidence of using unauthorized resources for completion.

Submitted code will be cross-checked for code that appears copied between students.

If you have any doubt with respect to what is acceptable to discuss, speak with the professor first.

ACADEMIC INTEGRITY AND THE USF HONOR CODE

Utilizing unauthorized materials or resources, such as those listed above, to complete course assignments, in whole or in part, is considered cheating and plagiarism and is therefore in violation of the USF Honor Code, which all students adhere to. If a student is suspected of cheating or plagiarism, an incident report will be filed with USF's Academic Integrity Committee to be kept on the student's record for the duration of their enrollment at the University. In addition, the student will receive a zero grade on the assignment in question. No exceptions.

EXTRA CREDIT

Throughout the course, extra credit assignments will be assigned and posted on Canvas. The completion of these assignments is optional. Failure to complete an extra credit assignment will not affect a student's grade. Completion of extra credit assignments awards a student extra points to be applied to the Homework Programs assignment category. Code submissions for these assignments are subject to the same Academic Honesty and Integrity policies as list above. Extra credit assignments are not accepted past their deadline and Assignment Extensions cannot be applied to them. Additional extra credit opportunities will not be given to individual students beyond what is posted on Canvas.

University Policies

CREDIT-HOUR POLICY

One unit of credit in lecture, seminar, and discussion work approximates one hour of direct faculty instruction (or 50 minutes plus a break) and a minimum of two hours of out-of-class student work per week through one 15-week semester. For further details, see USF's Credit Hour Policy.

STUDENTS WITH DISABILITIES

The University of San Francisco is committed to providing equal access to students with disabilities. If you are a student with a disability, or if you think you may have a disability, please contact Student Disability Services (SDS) at sds@usfca.edu or (415) 422-2613, to speak with a disability specialist. (All communication with SDS is private and confidential.) If you are determined eligible for accommodations, please request that your accommodation letter be sent to me as soon as possible; students are encouraged to contact SDS at the beginning of the semester, as accommodations are not retroactive. Once I have been notified by SDS of your accommodations we can discuss your accommodations and ensure your access to this class or clinical setting. For more information please visit the SDS website: <https://www.usfca.edu/student-disability-services>.

BEHAVIORAL EXPECTATIONS

The University of San Francisco is committed to providing an environment consistent with the academic nature and core values of the institution in which students can participate in learning as a humanizing, social activity rather than a competitive exercise to support the full, integral development of each person and all persons, with the belief that no individual or group may rightfully prosper at the expense of others.

It is important you know what is expected of you academically and behaviorally through the published course syllabus, the Student Conduct Code, and other instructions provided by the instructor. Therefore, you are expected to uphold the following:

- Behave in accordance with the Student Conduct Code and other USF policies
- Refrain from disrupting the ability of fellow students to learn or the instructor's ability to teach. Examples of disruption include:
 - Cell phone or computer use that significantly or repeatedly distracts others
 - Coming to class late, leaving early, or excessively physically relocating oneself in the classroom
 - Speaking frequently without being called on
 - Yelling, cursing, or engaging in aggressive behavior

- When interacting online, communicate in a respectful fashion. This includes, but is not limited to:
 - Refraining from name calling, profanity, or typing in all capital letters
 - Sending multiple emails with one sentence
 - Avoiding rants or discussing non-relevant topics

Open discussion and disagreement are encouraged when done respectfully and in the spirit of academic discourse. There are also a variety of behaviors that, while not against a specific University policy, may create disruption in this course. Students whose behavior is disruptive or who fail to comply with the instructor may be dismissed from the class for the remainder of the class period and may need to meet with the instructor or Dean prior to returning to the next class period. If necessary, referrals may also be made to the Student Conduct process for violations of the Student Conduct Code.

ACADEMIC INTEGRITY

As a Jesuit institution committed to *cura personalis*—the care and education of the whole person—USF has an obligation to embody and foster the values of honesty and integrity. USF upholds the standards of honesty and integrity from all members of the academic community. All students are expected to know and adhere to the University's Honor Code.

- Plagiarism — intentionally or unintentionally representing the words or ideas of another person as your own; failure to properly cite references; manufacturing references.
- Working with another person when independent work is required.
- Submission of the same paper in more than one course without the specific permission of each instructor.
- Submitting a paper written by another person or obtained from the internet.

COUNSELING AND PSYCHOLOGICAL SERVICES

Many college students experience mental health difficulties. Counseling and Psychological Services (CAPS) is a great source of support for addressing a range of mental health needs, including anxiety, stress, loneliness, relationship struggles, grief/loss, adjusting to change, identity development, racial/cultural concerns, and mild to moderate depression.

CAPS offers in-person and remote individual and group therapy to enrolled students. Students seeking services are scheduled for a 15-20-minute phone screening to assess immediate risk, as well as identify initial treatment needs, and make recommendations for next steps. These may include a crisis or risk assessment, brief individual therapy (every 2-3 weeks), single session therapy, weekly group therapy, or referrals to off-campus providers. Please note, CAPS does not have a psychiatrist on staff to prescribe medication. Students in need of medication or with more complex mental health concerns are referred off-campus for specialized, comprehensive, treatment and/or ongoing, longer-term therapy. To make an appointment, call CAPS at 415.422.6352 or request an appointment via the CAPS website. CAPS does not accept walk-in appointments.

Students with urgent mental health needs can call the CAPS All-Hours line (855-531-0761) to immediately speak with a mental health clinician. The CAPS All Hours line is available 24/7, including weekends/holidays, and accepts international calls, and should only be accessed for mental health crises. For all other mental health concerns, including scheduling, canceling, rescheduling, or leaving a message for a CAPS clinician, please call CAPS at 415.422.6352 during normal business hours.

All students are encouraged to check out the CAPS website and access our extensive online resources, podcasts, mental health apps, videos, self-care strategies, and more.

If you are concerned about a student and would like someone to follow up, please contact the Dean of

Students Office at 415.422.5330. If you know someone who is an immediate risk of harming themselves or others, please contact Public Safety at 415.422.2911 in San Francisco, out of state or additional locations within California dial 911, or call the National Suicide & Crisis Lifeline by dialing 988.

TITLE IX

The Title IX Office seeks to stop, remedy, and prevent occurrences of sex and gender-based discrimination, sexual harassment, and sexual violence. The University has a Policy on Nondiscrimination Based on Sex and Gender, Sexual Harassment, and Sexual Misconduct for all Students, Employees, and Third-Parties. If you have experienced any behaviors related to this policy, we encourage you to report the incident. If you report these behaviors to any staff or faculty member, they must notify the USF Title IX Coordinator.

Students who wish to report any sexual misconduct should use the online mandatory reporting form. You can also contact the Title IX Office directly by calling (415)422-4563, emailing titleix@usfca.edu, or visiting our office in Lone Mountain, Room 145.

As an employee at USF, and your Professor, I am a mandatory reporter, meaning I have to share any instances of sexual harassment or sexual violence shared with me or that become known to me. I will have to share this information, including names and any details known, to the Title IX Office to connect you with resources. If you would like more information about the resources available, you can ask me at any time this semester. You do not need to tell me why you are asking to get help for a friend, another student, or yourself.

If you are interested in accessing confidential resources, please consider these options. If you report any information to the following, it will not generate a report to the Title IX office:

- Students may speak to someone confidentially by contacting Counseling and Psychological Services at (415) 422-6352, or speaking to a clergy member in University Ministry at (415) 422-4463.
- To find more about reporting sexual misconduct (including sexual assault, relationship violence, or stalking) and resources at USF, visit the USF Title IX website.
- For off-campus resources, contact San Francisco Women Against Rape (SFWAR) at (415) 647-7273; www.sfwar.org

If during the course of the semester you need accommodations related to pregnancy or parenting, please let me know and I can connect you to the Title IX office. They are able to offer support through arranging accommodations, leave, or other support. The Title IX office is responsible for ensuring pregnant and postpartum students get the changes they need to stay healthy while being able to continue their education.

CONFIDENTIAL RESOURCES FOR REPORTING SEXUAL MISCONDUCT

Students may speak to someone confidentially which will not generate a report to the Title IX Office by contacting Counseling and Psychological Services at (415) 422-6352 during M-F 9-4pm, or speaking to a clergy member in University Ministry at (415) 422-4463.

If you need to speak to a mental health clinician immediately, please call the CAPS 24/7 All Hours Line at 855-531-0761 (available daily, including weekends and holidays, and accepts international calls), Public Safety (415-422-2911), 911, the Suicide Hotline (dial 988), or go to your nearest emergency room

For off-campus resources and local Bay Area organizations check here: <https://myusf.usfca.edu/title-ix/resources>

LEARNING, WRITING, AND SPEAKING CENTERS

The Learning, Writing, and Speaking Centers (LWSC) at USF provide individualized support to assist students in better understanding course material and to aid them on their path to success. Services are free and include tutoring, collaborative peer support services, academic skills coaching, writing, and speaking support. Services are available in-person and on Zoom.

LWSC staff can be reached Monday through Thursday between 9:00am-8:00pm and Friday between 9:00am-5:00pm at LWSC@usfca.edu or through the chatbox on our myUSF webpages or by phone at (415) 422-6713. To make an appointment for subject tutoring, academic skills coaching, the writing center, or the speaking center, students should visit the Student Appointment Dashboard.

COMMUNICATION

All course communications, like all other USF communications, will be sent to your USF official email address. You are therefore strongly encouraged to monitor that email account.

GLEESON LIBRARY

Looking for help with a research paper or project? Set up a consultation with a Librarian or get 24/7 research help online.

USF FOOD PANTRY

The USF Food Pantry is an intermediate, short-term solution for any registered USF student to receive food and toiletry resources. Students are invited to stop by the pantry located on the first floor of Gleeson Library in the Atrium, and take the items that they need. Items are available on a first-come, first-serve basis until our supply is depleted. You will be asked to check-in via QR code before entering the pantry. For more information and the current schedule, visit the USF food pantry website. If you have further questions, please contact the Pantry Coordinator at usfpantry@usfca.edu or 415-422-4099 (during business hours Monday thru Friday from 9:00am - 5:00pm). You can find out about additional food security resources through the USF food insecurity resource page and the CalFresh resources site.